

Chapter

Chemical Coordination in Plants

1. Stems are : SQP 2022

- (a) positively geotropic and negatively phototropic
- (b) positively geotropic and positively phototropic
- (c) negatively phototropic and negatively geotropic
- (d) negatively geotropic and positively phototropic

2. When roots of a plant always move towards water, even if it means going against the pull of gravity This indicates SQP 2023

- (a) hydrotropism
- (b) negative geotropism
- (c) geotropism
- (d) positive geotropism

3. Rohit wants to measure the growth in length of a plant for his project. The device used for this is: MAIN 2023

- (a) Sphygmomanometer
- (b) Manometer
- (c) Auxanometer
- (d) None of these

4. Define: COMP 2023

- (i) Phototropism.
- (ii) Hydrotropism.

5. Apical meristem, Positive phototropism, Auxins, Cell elongation. SQP 2025

6. The tropic movement of plant parts in response to chemicals. MAIN 2024

7. The movement of plant parts towards the stimulus of chemical is _____.
COMP 2023

8. Which parts of a plant exhibit: SQP 2025

(a) Negative Geotropism

(b) Positive Hydrotropism

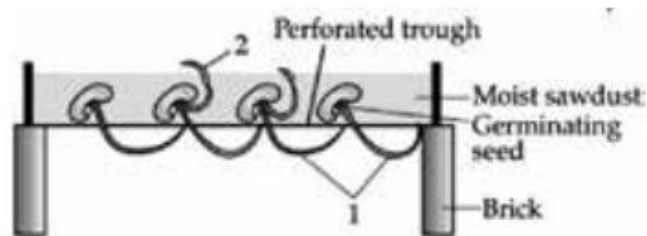
9. Give example of the following: COMP 2021

(i) Chemotropism

(ii) Thigmotropism

(iii) Phototropism

10. Given below is an experimental setup to demonstrate a particular tropic movement in germinating seeds. Study the diagram and answer the questions that follow : COMP 2022



(i) Label the parts 1 and 2.

(ii) Name the tropic movement shown by part 1.

(iii) Part 1 is affected by two stimuli. Name them. Which one of the two is stronger ?

(iv) What is Thigmotropism ? Give one example.

(v) What is meant by 'Positive' and 'Negative' tropic movements in plants ?

Solution

1. (d) Negatively geotropic and positively phototropic.

Stems grow upward, away from gravity (negatively geotropic) and towards sunlight (positively phototropic).

2. (a) Hydrotropism

Roots show hydrotropism as they grow towards moisture or water, even against gravity.

3. (c) Auxanometer

An auxanometer is an instrument used to measure the increase in length of a plant organ.

4. (i) **Phototropism:** Movement of plant parts towards or away from light, e.g. bending of stem towards light.

(ii) **Hydrotropism:** Movement of plant parts towards water, e.g. roots growing towards moisture.

5. Auxins → Apical meristem → Cell elongation → Positive phototropism.

6. Chemotropism

7. Chemotropism.

8. (a) Negative Geotropism – Shoot

(b) Positive Hydrotropism – Root

9. (i) **Chemotropism** – Growth of pollen tube towards the ovule

(ii) **Thigmotropism** – Growth of tendril around a support

(iii) **Phototropism** – Growth of stem towards light

10. (i) Label the parts:

- 1 → Radicle

- 2 → Plumule

(ii) **Tropic movement shown by part 1:** Hydrotropism

(iii) **Stimuli affecting part 1:** Gravity and Water

- **Stronger stimulus:** Water

(iv) **Thigmotropism:** Growth movement of a plant in response to touch, e.g., Sweet Pea tendril wrapping around a support.

(v) **Positive and Negative tropic movements:**

- **Positive:** Movement of plant parts **towards** the stimulus
- **Negative:** Movement of plant parts **away from** the stimulus